

CHANGWEI YAO

✉ chadwickyyu@outlook.com | 📞 213-500-7604 | 🌐 www.chad-yao.com | 🌐 changwei-yao

EDUCATION

Carnegie Mellon University (CMU)

M.S. in Information Systems

Pittsburgh, PA

Aug. 2024 - May. 2026 (expected)

- **Coursework:** Mobile Manipulation, Introduction to Robot Learning (Reinforcement Learning), Learning for 3D Vision

Hunan University (HNU)

B.E. in Information Security (Ranking: 1/41)

Changsha, China

Sep. 2019 - Jun. 2023

TECHNICAL SKILLS

Programming & Libraries

Python, C/C++, MATLAB, Bash, JavaScript, OpenCV, PCL

Software & Tools

Linux, Docker, Git, CMake, GDB, tmux, SSH, OpenClaw

Physical Robots

KUKA LBR iiwa7, Hello Robot Stretch 3, ARX X5, myCobot 320

Robotics & Simulation

ROS/ROS2, MuJoCo, Isaac Gym, Gazebo, PyBullet, MoveIt

PROFESSIONAL EXPERIENCES

CMU College of Engineering, Research Intern | Robot Learning, Reinforcement Learning, VLA

Sep. 2024 - Aug. 2025

CyLab Biometrics Center, Advisor: Marios Savvides

CMU, PA

- Built an automated RL reward design system leveraging Graph-of-Thoughts and vision-language models for long-horizon robotic tasks.
- Designed a bi-level optimization pipeline integrating LLM-based task decomposition with VLM-based rollout evaluation, eliminating human-in-the-loop reward tuning and improving task success rates by +32.25% on RoboGen while achieving 93.73% success on ManiSkill2.
- Developed a unified post-training framework for vision-language-action models using context-aware meta co-training to improve cross-task generalization and training efficiency.

CMU Robotics Institute, Research Intern | Interactive Navigation, Semantic Mapping

Sep. 2024 - Mar. 2025

Zhang's Lab, Advisor: Ji Zhang

CMU, PA

- Enabled real-time open-vocabulary semantic mapping by integrating LiDAR-based SLAM with vision-language perception, maintaining long-horizon instance consistency under occlusions and scene changes via a spatio-temporal tracking and update pipeline for dynamic objects.
- Developed a queryable 4D scene graph representation enabling language-guided navigation and spatio-temporal reasoning in the real world.

SUSTech Dept. of AIM, Research Staff | Imitation Learning, Distillation, Data Augmentation

Jul. 2023 - Jun. 2024

Control & Learning for Robotics and Autonomy Lab, Advisor: Wei Zhang

SUSTech, China

- Deployed an efficient robot control pipeline on the KUKA LBR iiwa7 by applying Consistency Models to Diffusion Policies, compressing multi-step inference into a few sampling steps and achieving 10x+ speedup for real-time, low-latency visuomotor control.
- Proposed a trajectory-level data augmentation method that minimizes interpolation between demonstration segments, improving synthetic data quality and increasing policy success rates by 4.8% and data generation success by 12.8% on MimicGen benchmarks.

HNU Dept. of CSEE, Undergrad | IoT Security, User Authentication, EM Sensing

Oct. 2021 - Aug. 2022

IoT Sensing and Security Lab, Advisor: Wenqiang Jin

HNU, China

- Built a zero-cost authentication system for heterogeneous IoT devices using human-induced EM coupling and rhythm-based signal patterns, achieving 98.67% accuracy in real-world deployment without requiring additional sensors or specialized hardware.
- Enabled robust cross-device authentication by extracting user-specific temporal EM signal features, improving resilience to environmental noise and spoofing attempts in heterogeneous IoT settings.

SELECTED PROJECTS

Whole-Body Mobile Manipulation 🤖: Learned a whole-body mobile manipulation policy without expert demonstrations by combining a GPU-vectorized whole-body controller, local reinforcement learning, and visual distillation for grasping diverse objects in cluttered scenes.

HiWE for Imitation Learning 🧑: Developed a semi-supervised imitation learning data processing pipeline on the ARX X5 with hierarchical waypoint extraction, improving policy success rates by 30–50% on real-world manipulation tasks using only 10% labeled data.

Deep-RL for Minecraft 🎮: Built a demonstration-driven Deep Q-Learning pipeline for MineRL tree-chopping, including dataset preprocessing, action-space construction, prioritized replay, and end-to-end training/evaluation in sparse-reward environments.

4D Human Reconstruction 🧑: Built a monocular 4D human reconstruction pipeline combining 3D tracking with generative augmentation to produce temporally consistent human representations, enabling robust reconstruction under occlusions and complex human motion.

SELECTED PUBLICATIONS

► **Changwei Yao**, X. Liu, C. Li, M. Savvides. "Reward Evolution with Graph-of-Thoughts: A Bi-Level Language Model Framework for Reinforcement Learning." *2026 IEEE International Conference on Robotics and Automation (ICRA)* (paper 📄)

► S. Zhao, G. Chen, H. Zhu, Z. Li, **Changwei Yao**, N. Zantout, S. Kim, W. Wang, J. Zhang, S. Scherer. "SuperMap: Spatio Temporal Semantic SLAM Enabling Robots to Understand Evolving World in Real Time." *2026 Proceedings of Robotics: Science and Systems (RSS)*. (demo 🎥)